



Bharat Heavy Electricals Limited

Powering Progress, Brightening Lives Touching Every Indian Home

SUBMITTED BY:

ADITYA ROHILLA

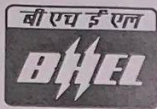
DEPARTMENT OF MECHANICAL ENGINEERING

(Specialisation in Automotive)

DELHI TECHNOLOGICAL UNIVERSITY

SUMMER TRAINING REPORT [8 WEEKS]

**STUDY OF MECHANICAL
AUXILIARIES REQUIRED BY BHEL IN
A THERMAL POWER PLANT**



ACKNOWLEDGEMENT

I, Aditya Rohilla, express my sincere gratitude to Bharat Heavy Electricals Limited (BHEL), Noida—one of India's premier engineering and manufacturing organizations—for granting me the invaluable opportunity to undertake an eight-week internship in the Mechanical Auxiliaries (MAUX) Department at the Project Engineering Management (PEM) office, Noida, Uttar Pradesh.

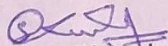
I extend my respectful thanks to Sh. Lalit Jindal (MAUX), Sh. Mohit Sharma (MAUX), Sh. Sachin Chauhan (MAUX), and Sh. Rajender Kumar (HR) for their guidance, encouragement, and constructive support throughout the duration of my training. I am equally grateful to all members of the MAUX Department for their cooperation, technical insights, and facilitation of my learning experience.

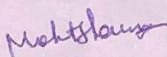
Finally, I convey my thanks to all individuals who, directly or indirectly, contributed to my internship and enhanced its academic and professional value.

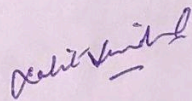
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INTRODUCTION

Bharat Heavy Electricals Limited (BHEL), founded in 1964, is India's preeminent engineering and manufacturing powerhouse, instrumental in propelling the nation's industrial and infrastructure growth.

With an unparalleled legacy spanning over six decades, BHEL has been pivotal in establishing India as a global leader in power generation, electrification, transportation, and strategic sectors. BHEL has supplied over 200 GW of power equipment, supporting more than 1,000 power plants across India and around the world.

Remarkably, BHEL contributes to 54% of India's thermal, 48% of nuclear, and 44% of hydroelectric power capacity, forming the backbone of the national power grid. On the global stage, BHEL-manufactured plants operate in 21 countries with a cumulative overseas installed capacity exceeding 9,000 MW.

The company's innovation portfolio boasts the world's first 1200 kV UHVAC transformer, advanced DC train engines with regenerative braking, and cutting-edge floating solar plants.

BHEL has produced more than 800 locomotives, 750,000 MVA in transmission systems, and delivered energy solutions on six continents. Its most recent fiscal year saw record-breaking order inflows, securing its status as a champion of "Make in India" and self-reliant national progress.

BHEL unequivocally remains the lifeblood of India's industrial ambition and technological excellence.



OBJECTIVE OF INTERNSHIP

As an intern in the Mechanical Auxiliaries department at Bharat Heavy Electricals Limited (BHEL), Noida, my primary objective was to immerse myself in the power generation domain of a Navratna PSU and translate theoretical knowledge into practical competency. By engaging with complex electrical and manufacturing systems, I aimed to strengthen my understanding of the latest practices and rigorous standards that align with BHEL's vision of being a client-focused, globally competitive energy enterprise.

I was particularly keen to explore the functioning of power plants and understand how they operate beyond theoretical concepts. During the course of the internship, I realized that power plants work within highly sophisticated and integrated systems, far more intricate than what is often perceived through academic study. My objective was to gain an in-depth, holistic perspective of these systems, contribute to in-house R&D initiatives, and learn from experienced engineers through collaborative project work.

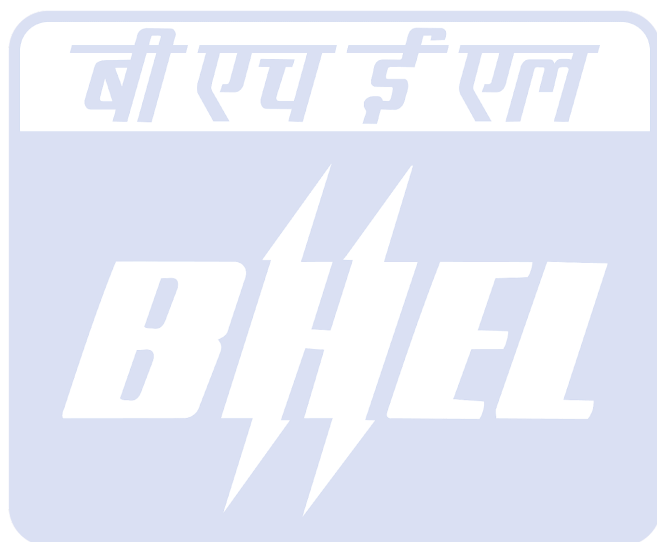
While I aspired to build comprehensive knowledge and contribute innovative ideas in this sector, the eight-week duration posed a challenge in fully realizing these ambitions. Nonetheless, the internship served as a valuable foundation for developing best practices in teamwork, engineering excellence, and continuous learning, which I intend to carry forward in my professional journey.

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ORGANISATIONAL STRUCTURE

Bharat Heavy Electricals Limited (BHEL) stands as India's premier engineering and manufacturing company in the energy and infrastructure sectors, recognized for its technological prowess and global presence.

- **Governing Body:** BHEL is overseen by a Board of Directors comprising Executive Directors (including the Chairman & Managing Director), Government Nominee Directors, and Independent Directors. This structure ensures both robust management and independent oversight, with 16 sanctioned Board positions—6 Functional Directors (including CMD), 2 Government Nominees, and 8 Independent Directors
- **Corporate Committees:** A 3-tier corporate committee system manages project reviews and corporate governance, ensuring efficient decision-making and policy adherence at multiple levels.
- **Decentralized Operations:** While administration is decentralized across divisions and regional offices, policy decisions remain centralized, supporting both operational agility and unified strategic vision.

Divisions and Spread

- **Manufacturing Powerhouse:** BHEL now operates 16 advanced manufacturing divisions and 8 specialist service centers across India, producing over 180 categories of products for diverse sectors.
- **Regional Reach:** The company manages 4 Power Sector Regional Centers, 4 regional offices, 9 service centres, and 15 regional marketing centres, ensuring thorough customer engagement and after-sales service.
- **Specialised Units:** These include heavy electrical plants, transformer plants, electronics divisions, R&D centres, and more—each contributing to sectoral expertise and rapid innovation.

Recent Achievements (2025)

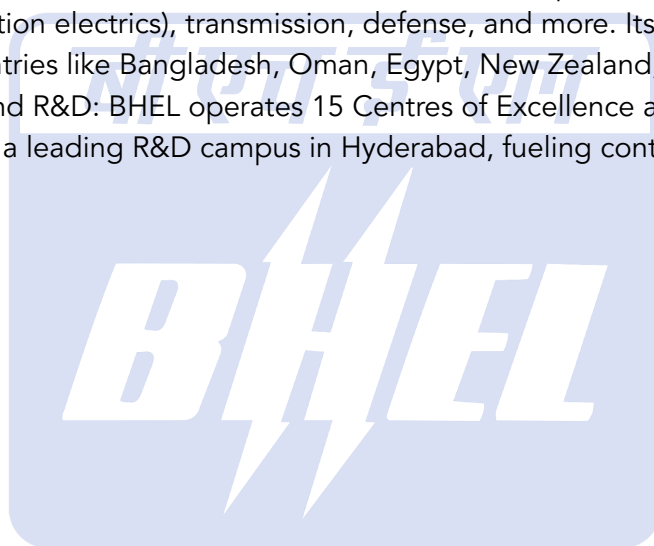
- **Revenue Growth:** For FY2024-25, BHEL reported a 19% increase in revenue, touching ₹27,350 crore, buoyed by a record order book of ₹92,534 crore. The company remains a clear market leader with over 8.1GW of power capacity delivered in the year, and a robust backlog of projects in execution.



- Awards & Recognition:
 - Winner of ICMAI Excellence in Cost Management (2024)
 - Honored with the EEPC India Export Excellence Award (2025)
 - Named Best PSU in overall HR Excellence, Best R&D Initiative, and Excellence in Learning and Development (Governance Now PSU Awards 2025)
- Employees: BHEL employs around 28,000 highly skilled workers, making it one of the largest employers in Indian engineering.

Global Footprint

- International Presence: BHEL exports to over 90 countries across all six continents, with equipment installed in iconic infrastructure projects worldwide—from power plants to industrial systems.
- Major Sectors Served: Power generation, industry, transportation (notably Indian Railways traction electrics), transmission, defense, and more. Its integrated solutions support countries like Bangladesh, Oman, Egypt, New Zealand, and many more.
- Innovation and R&D: BHEL operates 15 Centres of Excellence and 5 research centers, with a leading R&D campus in Hyderabad, fueling continual technological upgrades.





Vision

A world-class innovating, competitive, and profitable engineering enterprise providing total business solutions.

MISSION

To be the leading Indian engineering enterprise providing quality products systems and services in the fields of energy, transportation, infrastructure, and other potential areas.



VALUES

- Meeting commitments made to external & internal customers
- Foster learning creativity and speed of response
- Respect for dignity and potential of individual
- Loyalty and pride in the company
- Team playing

BHEL's vision is to become a world world-class engineering enterprise, committed to enhancing stakeholder value. The greatest strength of BHEL is its highly skilled and committed employees. Spread all over India & abroad to provide prompt and effective service to customers.

BUSINESS SECTOR

BHEL operations are organized around business sectors to provide a strong market orientation. These business sectors are Power Indus and International operations.

POWER SECTOR

BHEL's offerings include:

- ❖ Steam turbines, generators, boilers, and matching auxiliaries for fossil-fuel applications up to 1000 MW unit size.
- ❖ Emission control equipment including Flue Gas Desulphurization systems for SO_x emission control, high-efficiency Electrostatic Precipitators for particulate emission control, and Boiler modification and Selective Catalytic Reduction systems for NO_x emission control.
- ❖ Gas turbines and generators up to 299 MW unit size.
- ❖ Hydro turbines and generators up to 400 MW unit size.
- ❖ 220/235/500/540/700 MWe nuclear turbine generator sets.
- ❖ Plant performance improvement through renovation, modernization, flexibilization, uprating, residual life assessment, health diagnostics, and life extension of power Plants.



INDUSTRY SECTOR

BHEL is a major contributor of equipment and systems to important industries like:-

- ❖ Cement
- ❖ Petrochemicals
- ❖ Fertilizers
- ❖ Steel paper
- ❖ Refineries
- ❖ Mining and Telecommunication

TRANSPORTATION

BHEL supplies a wide range of equipment and systems to Indian Railways.

- ❖ Electric locomotive
- ❖ Traction electric and traction control equipment

TELECOMMUNICATION

BHEL also caters to the Telecommunication sector by way of small, medium, and large switching systems.

DIVISIONS/UNITS OF BHEL

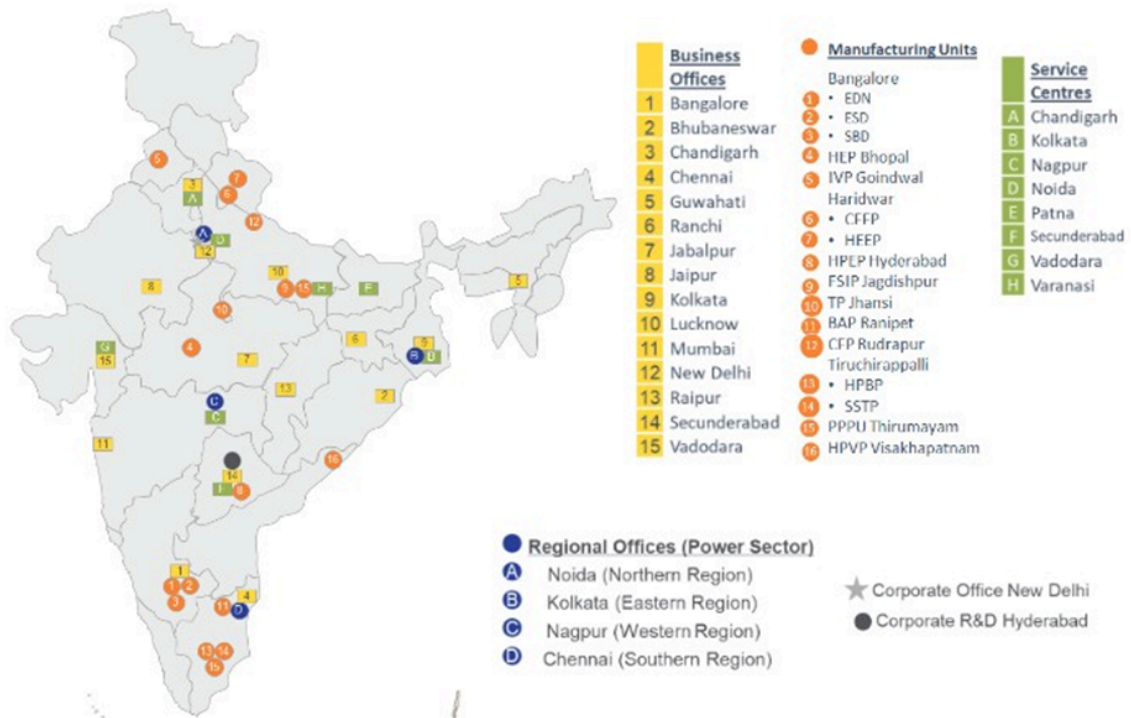
BHEL Manufacturing Units	Bengaluru	Electronics Division (EDN)
		Electronics Systems Division (ESD)
		Solar Business Division (SBD)

	Bhopal	Heavy Electrical Plant (HEP)
	Goindwal	Industrial Valves Plant (IVP)
	Haridwar	Heavy Electrical Equipment Plant (HEEP)
		Central Foundry Forge Plant (CFFP)
	Hyderabad	Heavy Power Equipment Plant (HPEP)
	Jagdishpur	Fabrication Stamping Insulator Plant (FSIP)
	Jhansi	Transformer Plant (TP)

	Rudrapur	Component Fabrication Plant (CFP)
	Ranipet	Boiler Auxiliaries Plant (BAP)
	Tiruchirappalli	High Pressure Boiler Plant (HPBP)
		Seamless Steel Tube Plant (SSTP)
	Thirumayam	Power Plant Piping Unit (PPPU)
	Visakhapatna m	Heavy Plates & Vessels Plant (HPVP)



BHEL Repair Units	Mumbai	Electrical Machine Repair Plant (EMRP)
	Varanasi	Heavy Equipment Repair Plant (HERP)
BHEL Power Sector Regions	PSNR	Power Sector Northern Region
	PSER	Power Sector Eastern Region
	PSWR	Power Sector Western Region
	PSSR	Power Sector Southern Region
Engineering & Supply	Bengaluru	Industrial Systems Group (ISG)
	TBG	Transmission Business Group (TBG)
	Hyderabad	Project Engineering & Systems Division (PE&SD)
Other Units	Chennai	Piping Centre





PEM: The Nerve Center of BHEL Projects

PEM originated in 1974, first as the Consultancy Services Division, before transforming into its current form in 1984. Today, it's the *nerve center* that coordinates all technical functions required to deliver thermal, gas-based, and nuclear island projects, as well as interfacing with non-conventional energy solutions. It manages the engineering, procurement, construction (EPC), and commissioning processes—bridging customers, consultants, statutory bodies, BHEL's own manufacturing units, project sites, and vendors into one cohesive workflow.

Key Roles of PEM:

- System Integration: Ensures smooth cross-disciplinary handshakes—Civil, Electrical, Mechanical—right from design to commissioning.
- Relationship Building: Maintains lasting liaisons with national regulatory agencies, vendors, regions, and units.
- Quality Focus: Implements robust Quality Management Systems (QMS) that set standards for each engineering stream.

The MAUX Department: Where Every Power Plant System Connects

Core Segments

1. Heating, Ventilation and Air Conditioning (HVAC): Critical for comfort and safety, requiring knowledge of load calculations, contamination control, and fail-safes.
2. Plant Facility: Involves layout planning, inter-plant logistics, and utility systems that impact every discipline and department.
3. Flue Gas Desulphurisation (FGD): Mandated for emissions control, this segment merges chemistry, mechanical design, and environmental standards—requiring seamless collaboration with system and control engineering.
4. Material Handling: The lifeline for raw and finished goods—impacting everything from timelines to maintenance planning.



Cross-functional Connectivity

No department in PEM functions in isolation:

- Civil, Electrical, and Mechanical teams interact daily. For example, Mechanical Piping layouts determine civil foundation loads; electrical supply planning must match with motor loads and HVAC for control rooms.
- System and Control Engineering sits at the intersection—instrumentation for FGD, automation for HVAC, and plant-wide monitoring are all multilayered collaborations.
- Water System Engineering is key for cooling, raw water, and condensate—these intersect with chemical dosing (handled by MAUX), electrical pumping stations, and civil storage structures.

MAUX thus houses deep domain knowledge: mechanical, chemical, civil, environmental, and some electrical. Team members must understand how their choices (say, a change in material specification for a pump) ripple through electrical sizing, civil supports, system controls, and operational protocols.



My Internship Journey: -As a Mechanical Engineering student passionate about energy systems, my internship at BHEL's MAUX Department in Film City, Noida (June-August 2025) was a pivotal step in my career. Having previously studied advancements at Narora Nuclear Power Plant, I was eager to explore thermal power auxiliaries at BHEL, a leader with 53% of India's thermal capacity .

I started my internship wanting to understand the MAUX Department at BHEL deeply. Those first 4-5 days were a whirlwind of exploration—every workflow, process, and document was new, and there was so much to absorb. Genuinely, it set the base for everything I did after.

Key Moments:

- Location: BHEL Office, Film City, Noida – a vibrant hub for engineering innovation.
- First Week: Overwhelmed but excited, I spent 4-5 days absorbing MAUX's role in power plants.
- Mentors: Sachin Sir's insights made complex concepts accessible; Lalit Sir's permissions opened practical doors; Utkarsh Sir's specs clarified real-world applications; Rajender Kumar Sir taught me about BHEL's workflows and Rajbhasha Hindi committee.
- Fellow Interns: Aryan Jain (conclusions), Fizal (Psychrometric Chart, Mill Reject), Jasjyot Singh Talwar (layout/piping)
- Goal: Bridge theory (e.g., PK Nag's Power Plant Engineering) with practice, focusing on HVAC and plant efficiency.

MAUX Department Overview (Days 1-5)



The MAUX (Mechanical Auxiliary) Department at BHEL Noida designs and implements support systems for power plants, ensuring seamless operation of main generation units. These auxiliaries account for 5-10% of plant energy but are crucial for efficiency (e.g., HVAC maintains control room temperatures at 22-24°C to prevent equipment failures).

Detailed Overview:

- **Role in Power Plants:** Auxiliaries like HVAC, water treatment, and fire protection support core operations, reducing downtime by 15-20% .
- **BHEL's Contribution:** MAUX supports BHEL's 53% share in India's thermal power capacity (e.g., Korba West STPP FGD systems).
- **Learning Process:** Reviewed department documents and discussed with Sachin Sir on auxiliary integration.

System	Description	BHEL Example	Impact on Efficiency
HVAC	Climate control for plant areas	R-134a chillers in Yerams Thermal PP	Prevents 5-10% equipment loss due to overheating
Water Treatment	Demineralization and cooling	NTPC-BHEL 800MW AUSC Plant	Reduces scaling, saving 2-3% energy
Fire Protection	Sprinklers and CO2 systems	Chandrapur STPP	Ensures safety compliance, minimizing shutdowns
Compressed Air	Instrumentation and pneumatics	Ramagundam STPP	Supports controls, 1-2% overall savings
Mill Reject	Handling coal mill rejects	Korba West STPP	Reduces mill wear, 10% downtime cut

HVAC Overview



Stepping into the HVAC Segment

After that initial overview, I decided to focus on the HVAC segment. Here, I went through each crucial component—evaporators, condensers, compressors, expansion valves, and chillers. I spent dedicated time just grasping their theory and how each one fits into the bigger picture.

The best part? Sachin sir believed in guiding me at every step, always patient, always insightful. It's because of his mentorship that the theoretical parts genuinely clicked for me and I could relate them to practical aspects.

HVAC (Heating, Ventilation, and Air Conditioning) systems in power plants maintain optimal environments for equipment reliability and personnel comfort by:

- Preventing temperature stratification
 - Removing contaminated air and waste heat
 - Providing fresh outside air
 - Evacuating smoke and heat in emergencies
-

Key HVAC Terminology

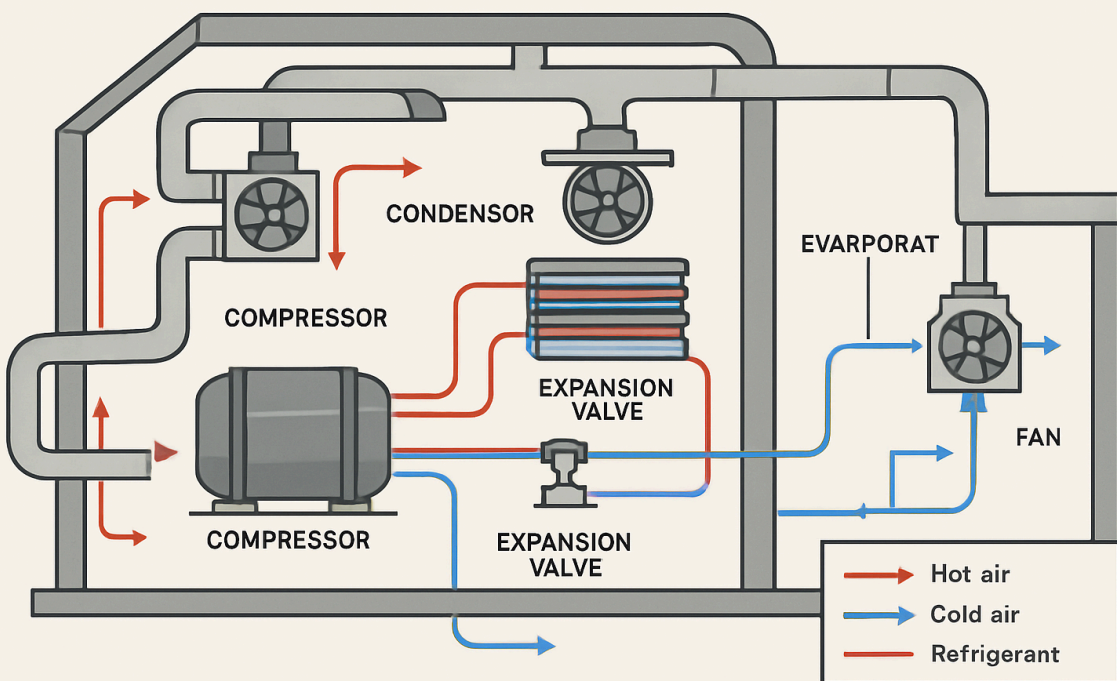
- Comfort Air Conditioning: Focus on human comfort in occupied spaces.
 - Process Air Conditioning: Focus on conditions for industrial processes; human comfort secondary.
 - Dry Bulb Temperature (DBT): Actual air temperature.
 - Wet Bulb Temperature (WBT): Temperature measured with a wetted sensor, important for humidity and cooling calculations.
 - Dew Point: Temperature where moisture condenses from air.
 - Relative Humidity (RH): % of moisture compared to max air can hold at that temperature.
 - Psychrometric Chart: Graphical tool to analyze air properties (temperature, humidity, enthalpy).
-

Major Components of HVAC Systems

- Compressor: Raises refrigerant pressure and temp in refrigeration cycle; types include screw, centrifugal, scroll, turbocor, etc. Modern compressors have VFDs and magnetic bearings for efficiency and reliability.
- Condenser: Removes heat from refrigerant vapor, condensing it to liquid. Can be air-cooled or water-cooled.
- Evaporator: Absorbs indoor heat by evaporating refrigerant; cools supply air.
- Expansion Valve: Controls refrigerant flow and pressure reduction to evaporator.
- Air Handling Unit (AHU): Circulates air through cooling coils and filters, distributing conditioned air via ducts.
- Fans/Blowers: Facilitate air movement in the system; their placement affects heat load calculations.

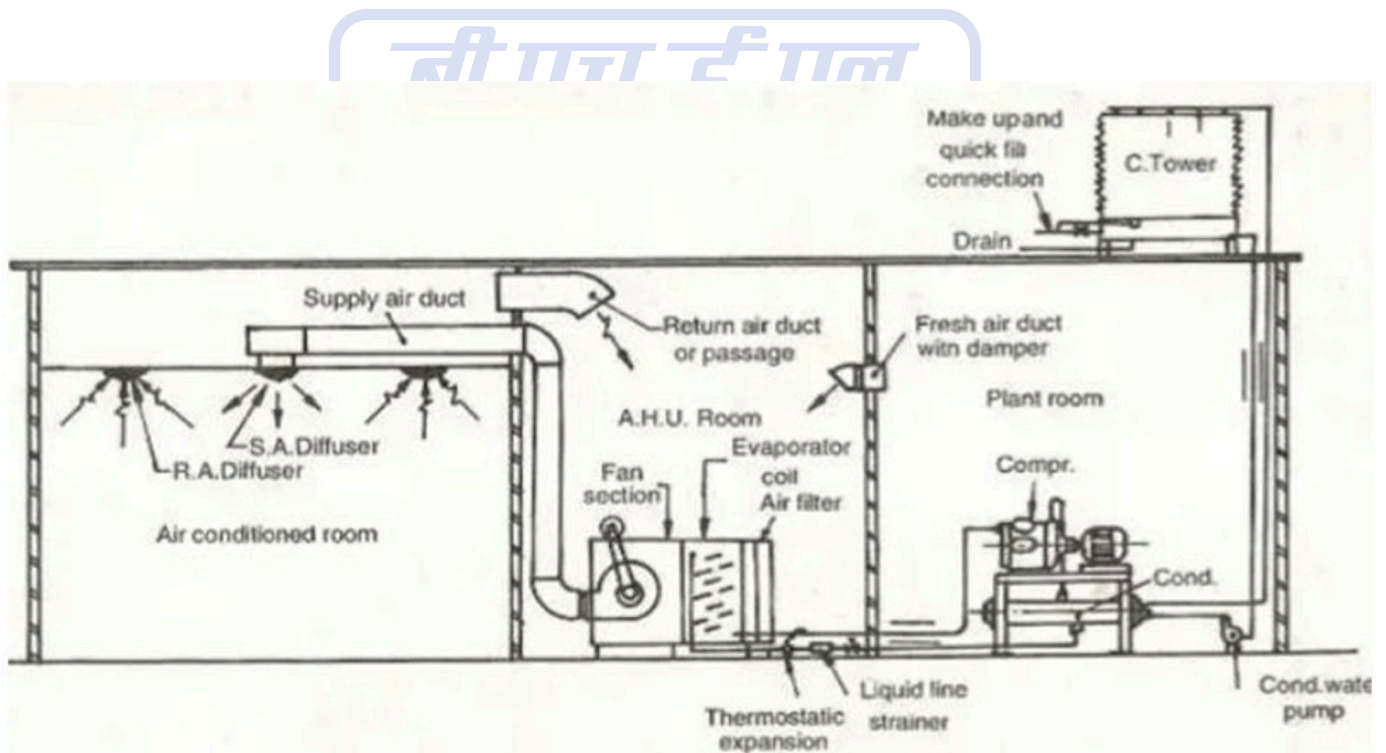
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POWER PLANT HVAC SYSTEM



HVAC Cooling Systems Types

- Direct Expansion (DX) Systems: Refrigerant directly cools air in AHU; efficient but limited by refrigerant piping length and building size. Suitable for smaller or single-floor buildings.
 - Chilled Water Systems: Use water chilled by a central chiller circulated to AHUs; suitable for large multi-story buildings with flexible layouts; slightly less efficient due to intermediary fluid but allow longer piping distance.
 - Vapor Absorption Chillers (VAM): Use heat sources (e.g., steam) instead of electricity to drive cooling, ideal where waste heat is available or electrical costs are high. They use water/LiBr solution as working fluid for eco-friendly operation.
-



Design Considerations & Load Calculations



- Loads account for heat gain from solar, conduction, internal equipment, occupants, lighting, and ventilation, including sensible and latent components.
 - Tools like Cooling Load Temperature Difference (CLTD) and Solar Cooling Load Factor (SCL) help estimate dynamic heat gains.
 - Diversity factors reduce total load based on actual occupancy and usage patterns.
 - Calculation of supply air volume depends on sensible heat load and desired inlet/outlet temperatures.
-

Advancements & Best Practices

- Use of Variable Frequency Drives (VFDs) on compressors and fans improves part-load efficiency dramatically.
 - Modern magnetic bearing compressors reduce maintenance and improve reliability.
 - Integration of IoT and Building Management Systems (BMS) allows for optimized HVAC operation, energy savings, and predictive maintenance.
 - Use of environmentally friendly refrigerants (e.g., R32, R290) and waste heat recovery technology enhances sustainability.
 - Advanced filtration and ventilation ensure high indoor air quality, crucial in plant environments for health and equipment longevity.
-

Diving into HVAC: Theoretical Foundations



After the department overview at Noida, I decided to focus on the HVAC segment, inspired by its critical role in maintaining optimal conditions for power plant operations. I spent the initial days studying evaporators, condensers, compressors, expansion valves, and chillers, gaining theoretical knowledge from resources like PK Nag's Power Plant Engineering and online videos.

Why HVAC? In power plants, HVAC ensures equipment reliability by controlling temperatures in high-heat environments (e.g., turbine halls at 45°C ambient need cooling to 25°C) and maintains operator comfort, saving 5-10% in energy losses from overheating (ASHRAE standards).

Sachin Sir's guidance was invaluable—he explained how HVAC acts as the "lungs of a power plant," preventing corrosion and failures in control rooms.



HVAC Components: Deep Dive and Insights from Sachin Sir



I delved into HVAC components through videos and Sachin Sir's practical insights, who blessed me with explanations that bridged theory and BHEL's real-world applications.

- **Evaporators:** Absorb heat from air/water, causing refrigerant evaporation. Types: Direct Expansion (DX) for control rooms (COP 3-4, used in BHEL's Noida designs), Flooded for large cooling loads (e.g., boiler areas). Material: Copper tubes with aluminum fins for corrosion resistance.
- **Condensers:** Release heat, condensing refrigerant. Types: Air-cooled for water-scarce sites, Water-cooled shell-and-tube (copper, 95% efficiency) preferred in BHEL projects like Yerams Thermal PP.
- **Compressors:** Increase refrigerant pressure/temperature. Types: Centrifugal (500 kW, BHEL's choice for 300-ton chillers), Screw for variable loads (energy savings 15-20% with VSD), Reciprocating for small auxiliaries.
- **Expansion Valves:** Reduce pressure for cooling. Types: Thermostatic (TXV, 0.5 bar drop, common in BHEL), Electronic for precise control in VRF systems (Daikin-inspired).
- **Chillers:** Central units; Vapor Compression (R-134a, 300-ton in BHEL) vs. Absorption (LiBr, waste heat-driven for 1-1.5 COP).

Sachin Sir highlighted how compressors in BHEL's systems use VSD for 20% energy reduction, a tip that enhanced my understanding.

Component	Type	Material	Efficiency	BHEL Application	Insights from Sachin Sir
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Evaporator	Flooded	Copper/Aluminum	COP 3-4	Turbine hall cooling	"Absorbs latent heat like a sponge—key for humidity control in Noida's humid climate."
Condenser	Shell-and-tube	Copper	95%	Yerams Thermal PP	"Water-cooled saves 10% energy in hot summers."
Compressor	Centrifugal	Steel	80-85%	300-ton chiller	"VSD is a game-changer for variable loads in power plants."
Expansion Valve	Electronic	SS316	N/A	VRF systems	"Precise like a surgeon—essential for efficiency."
Chiller	Vapor Compression	N/A	SEER 10-12	Ramagundam STPP	"R-134a is reliable, but we need greener options for future."

Practical HVAC Implementations



Wanting to see how things actually work, I approached Lalit sir for permission and then reached out to Utkarsh sir, who is my go-to expert for specs and practical insights in HVAC Systems . Under his guidance, I analyzed the specifications of HVAC systems used at BHEL, digging deep over two days to understand every last detail.

It took me approximately 2 days to analyze the specs, marking things for improvements.

Practical Specs as per BHEL :

- Chillers: 300-ton capacity, R-134a refrigerant, centrifugal compressor (500 kW, VSD-enabled for 15-20% energy savings).
- Evaporators: Shell and Tube type, chilled water outlet, copper tubes for high heat transfer.
- Condensers: Shell-and-tube, copper, 35C cooling water outlet, integrated with plant cooling towers.
- Compressors: Screw Type, 50 bar pressure rating.
- Expansion Valves: Electronic, 0.5 bar pressure drop, SCADA-integrated.
- System Integration: HVAC tied to plant SCADA (IEC 61850) for remote monitoring, used in BHEL's Yerams Thermal PP and Ramagundam STPP



Refrigerant Analysis and Proposed Improvements

Marking Opportunities for Improvement

Once I'd decoded the system, it was obvious that improvement was possible in various areas. One big area was the refrigerant section. I noticed BHEL uses R-134a refrigerant, which is definitely environment-friendly but, honestly, feels outdated now when compared to what the industry is moving towards. It also doesn't align with the net-zero targets or government's 2028 emission benchmarks.

Digging more, I found R-32 is the market's current best, and Daikin is pretty much the leader here. Of course, using R-32 needs replacing the chiller, which can push up installation costs. But for new setups, it beats R-134a in almost every way—cost, compliance, efficiency, and future-proofing.

Supporting Data

For a new 300-ton HVAC system (roughly 750kg refrigerant required):

- R32 (₹350/kg): Total cost = ₹262,500
- R134a (₹550/kg): Total cost = ₹412,500

That's about ₹150,000 saved up-front, plus lower electricity bills and fewer regulatory headaches over the years. R-32 just makes sense for new installs.

Yet, for older systems needing to meet norms, partial retrofit is possible. I found that R513A or R450A work well with minimal changes, lowering global warming potential by 50–60%.

This was the natural pause point in my refrigerant research.

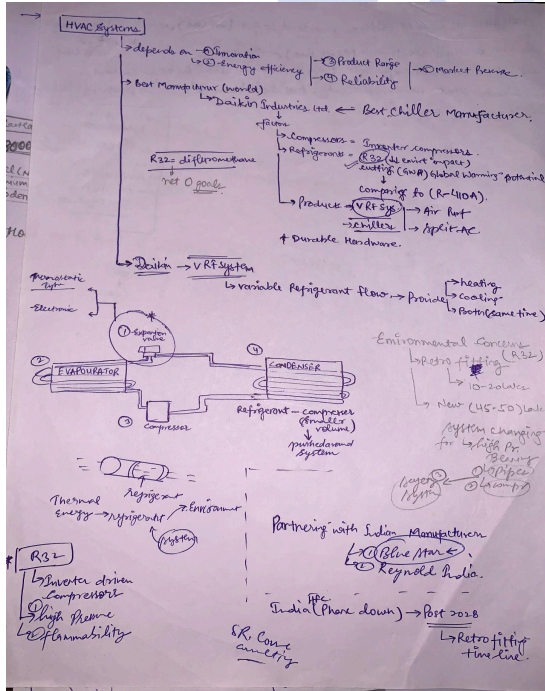
While analyzing specs, I marked improvement areas, focusing on refrigerants. BHEL uses R-134a (GWP 1430), which is environment-friendly but outdated—it neither meets net-zero

emissions targets nor the government's 2028 HFC phase-down under the Kigali Amendment (significant reductions post-2028).

Research showed:

- R-134a: High GWP, lower efficiency (SEER 10-12), facing phase-down risks (penalties post-2028).
- Proposed: R-32 (GWP 675) for new installations. Produced extensively by Daikin, R-32 is used in households (e.g., Daikin, LG, Voltas ACs—millions sold in India for residential cooling, 30% energy savings via inverter tech). For industry, we can take help from Indian manufacturers like Blue Star or Voltas to produce and install at scale, localizing Daikin's tech for BHEL's power plants.
- Retrofitting Old Systems: R-513A (GWP 631) or R-450A (GWP 601)—drop-in options with minimal changes (lubricant check, minor adjustments), reducing GWP by 50-60%.

Analysis :- R-32 achieves net-zero alignment by lowering GWP 75% vs. R-410A (similar to R-134a), and supports 2028 phase-down (India's gradual HFC reductions). Indian manufacturers like Blue Star (10-12% market share) can help produce R-32 chillers at {20-30%} lower cost than imports, enabling BHEL to install them in new projects.





Cost-Benefit Analysis of Refrigerants: Retrofitting vs. New Installations

For a 300-ton HVAC system (750 kg charge, common in BHEL power plants):

Refrigerant Cost Comparison

Refrigerant	Charge (kg)	Cost/kg (INR)	Total Refrigerant Cost (INR)	GW P	SEER	Annual Electricity Savings (INR lakh)	Notes
R-134a (Current)	250	550	1,37,500	1430	10-12	Baseline	High GWP, phase-down risk
R-32 (New Installs)	250	350	87,500	675	14-16	5-10 (15-20% reduction)	Household ACs use (Daikin/Voltas), industrial with Blue Star
R-513A (Retrofit)	250	600	1,50,000	631	10-12	0-1	56% GWP cut, minimal changes
R-450A (Retrofit)	250	580	1,45,000	601	10-12	0-1	58% GWP cut, lubricant compatible



Retrofitting Costs and Benefits (R-513A/R-450A):

- Cost: 5-10 lakh INR- chiller (lubricant flush, seal replacements, 1-2 lakh INR labor.
- Benefits: 50-60% GWP reduction, no full replacement, compliance with 2028 phase-down (avoid penalties 2-5lakh INR/plant/year. Electricity savings minimal (0-5%), but environmental credits 500-1000 INR/tonne CO2 20-40 crore INR/year for 400,000 tons avoided).
- Payback: 3-5 years via credits and reduced regulatory risks

New Installations Costs and Benefits (R-32):

- Cost: 262,500INR refrigerant + 50-75lakh INR system (inverter tech for households scaled to industry).
- Benefits: 15-20% energy savings 5-10lakh INR/year, $500 \text{ kW} \times 10,000 \text{ hours} \times 5 \text{ INR} \times \text{per} \backslash \text{kilo} \backslash \text{watt} \backslash \text{hour} \times 15\% = 7.5 \text{ lakh INR/year}$, lower GWP (75% cut vs. R-410A equivalent), meets net-zero goals. Household use (Daikin ACs: 30% efficiency, millions sold) scales to industry with Blue Star/Voltas help (localized production cuts costs 20-30%).
- Payback: 4-6 years, plus carbon credits.

Reaching Desired Goals:

- 2028 Phase-Down: R-32/R-513A meet Kigali targets (HFC reductions post-2028).
- Net-Zero: Reduces CO2 20-100% in HVAC (part of plant's 5-10% auxiliaries).
- Collaboration: Indian manufacturers like Blue Star (Noida facility, 10-12% market) or Voltas (household R-32 ACs) can produce/install at industry level, reducing BHEL's import dependency (20-30% costs). Partnership cost: 10-20 crore INR for tech transfer, yielding 50 lakh INR /plant savings.

[Analysis: (Daikin), R-32's household success (e.g., Daikin VRV ACs, 30% savings) translates to industry with Indian partners. Retrofitting achieves 50-60% GWP cut for legacy systems like Years PP, while new installs with R-32 align with net-zero by 2070.]

Exploring VRF Systems and Strategic Partnerships with Indian Manufacturers

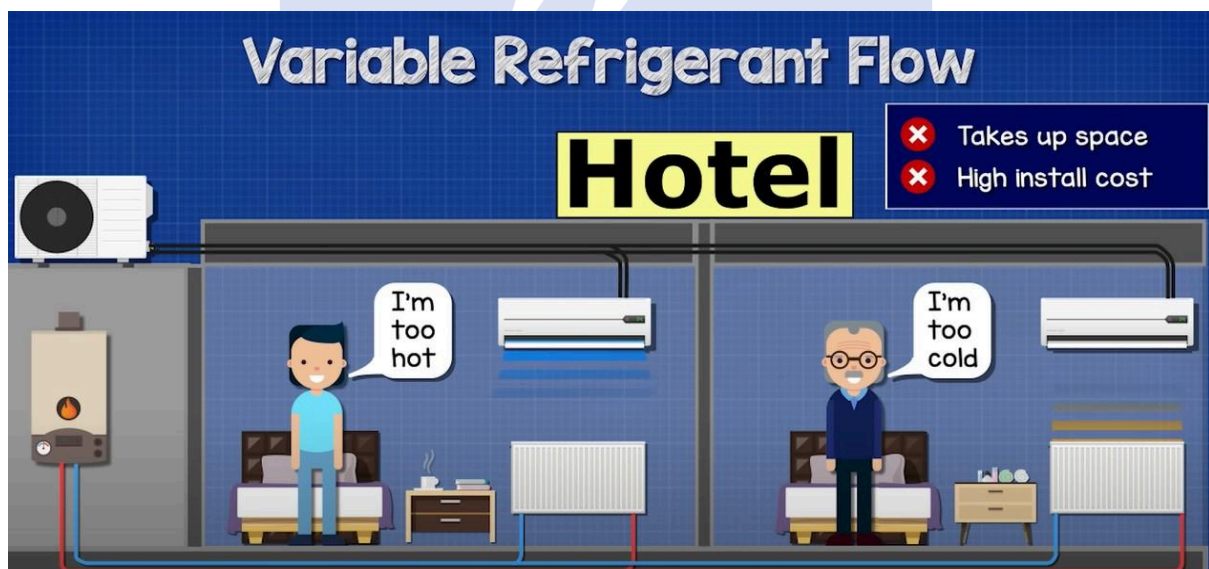
While on the refrigerant trail, I also explored Variable Refrigerant Flow (VRF) systems from Daikin, learning a lot through video breakdowns and experimenting with AI-based simulations. Expanding the net, I looked at how partnerships—both local (Blue Star, Reynold India) and global—could put BHEL ahead for the 2028 phase-down curve.

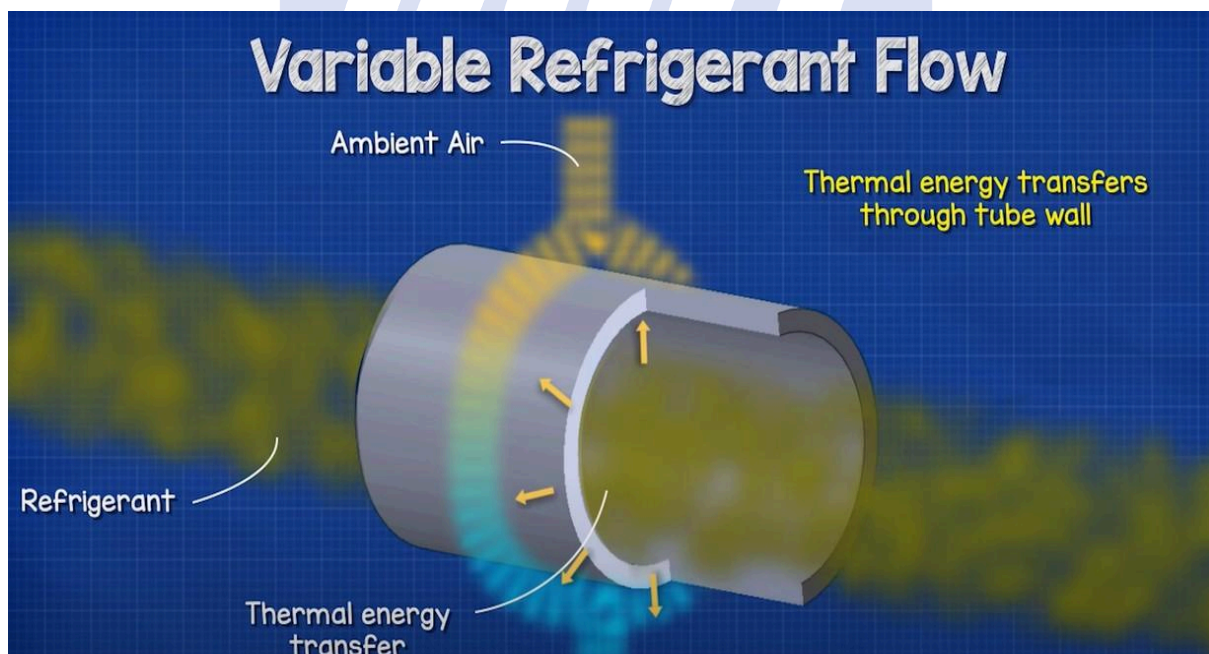
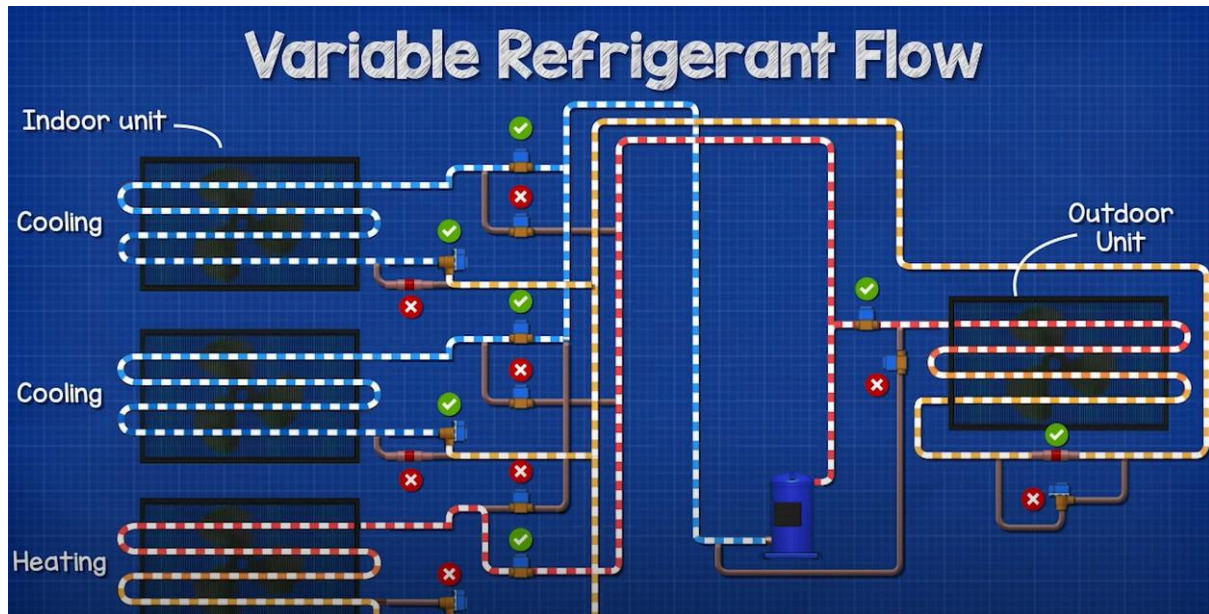
Studied Daikin's VRF systems via videos and AI: Multi-zone, inverter compressors, R-32, 30% energy savings vs. traditional HVAC.

BHEL Potential: Implement VRF in control rooms for zoned cooling, saving 20-25% energy.

Strategic Partnerships:

- Blue Star (Noida): 10-12% market share, MEP expertise; collaborate for industrial R-32 VRF production/installation 20-30 crore INR joint venture, 20% cost reduction).
- Reynold India: ASME-certified chillers, exports 60+ countries; localised R-32 for BHEL's new plants.
- Voltas: Household R-32 AC leader; scale to industry for retrofits/new installs, helping BHEL meet 2028 phase-down.







Transition to Power Plant Systems: A Broader Perspective

After pausing HVAC refrigerant research, I transitioned to overall power plant systems at BHEL Noida, motivated by a desire to understand efficiency holistically. I read PK Nag's *Power Plant Engineering* and watched videos like Kusile Power Station (<https://youtu.be/13Pb7HIB-oE?si=vN4OCJtTaAywFPWB>), the world's largest coal plant (4,800 MW, supercritical tech with FGD).

Kusile's real-time operations (boiler combustion, turbine spinning) contrasted with Nag's theory, showing practical challenges like soot buildup.

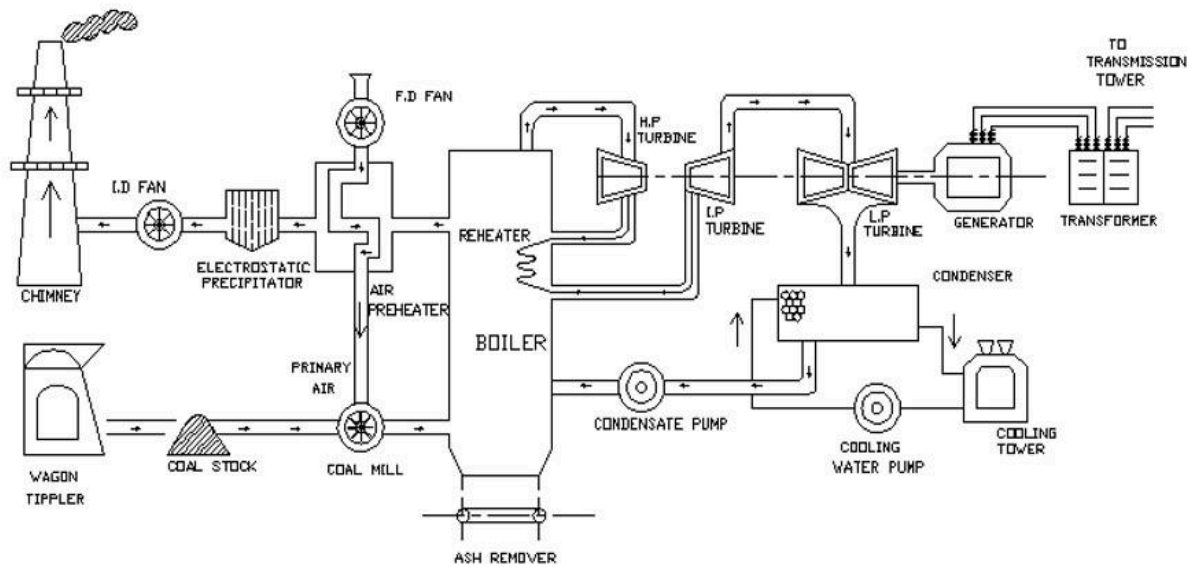


This shift felt natural after HVAC, echoing my study's focus on system-wide efficiency.

Sachin Sir's discussions on plant integration fueled my curiosity.

Analysis: Kusile's FGD (95% SO_x removal) aligns with BHEL's Korba West, but highlights waste heat losses (30-40%), leading to my molten salt exploration.

SCHEMATIC DIAGRAM



Fuel Cycle in Power Plants

Fuel Cycle: Coal receipt -> Handling (conveyors) -> Crushing (500 tph) -> Pulverization (50 tph mills) -> Combustion (boiler) -> Ash disposal (silos).

Efficiency Losses: 5-10% in pulverizers (friction heat), 2-5% in ash handling (dust emissions).

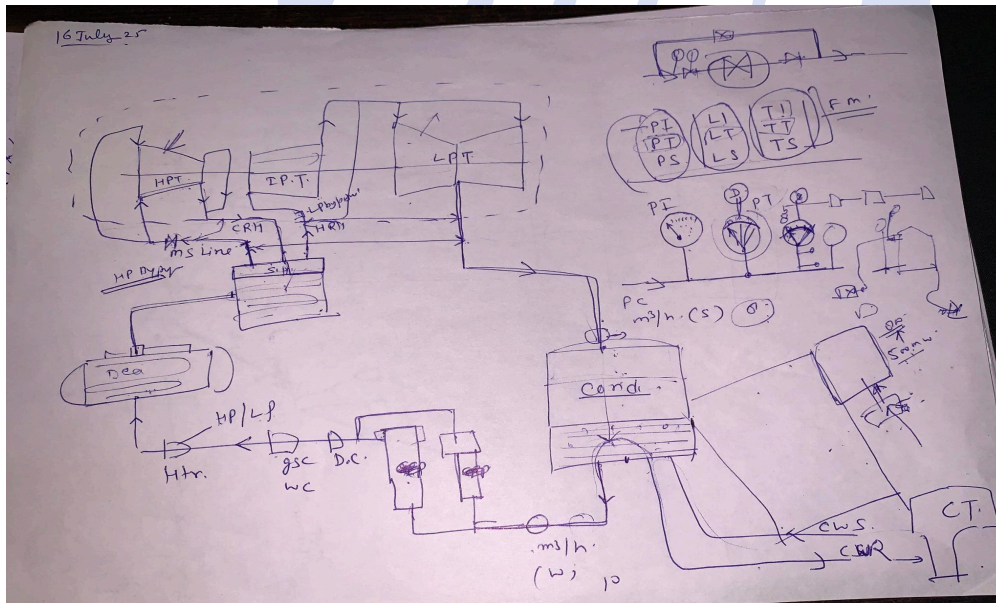
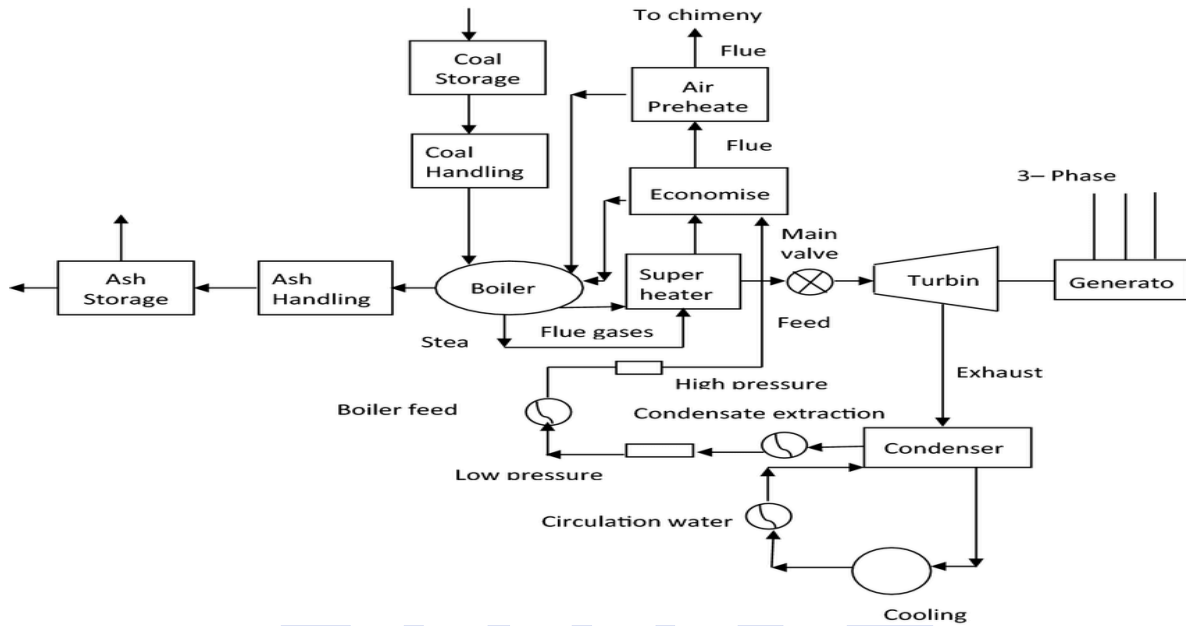




Table: Fuel Cycle Components and Losses

Stage	Equipmen	Loss (%)	BHEL Example	Improvement Suggestion
Pulverizer	Bowl mill	5-10	Ramagundam STPP	Biomass co-firing (10-20% CO2 cut)
Ash Handling	ESP + Silo	2-5	Korba West STPP	Double sprinkling (reduces dust 50%)

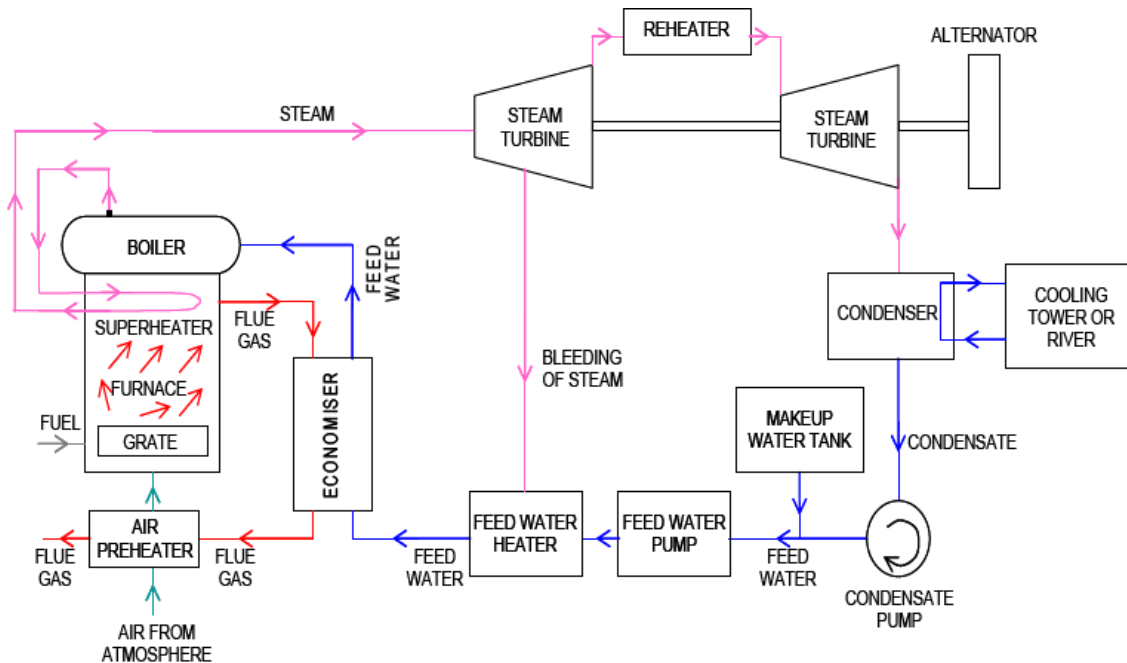
Analysis: From Kusile video and past searches (CoalHandlingPlants.com), biomass co-firing (Drax UK model) can reduce CO2 10-20%.

Analyzing losses reminded me of Narora's fuel handling optimizations; Fizal's MRHS insights tied in perfectly here.

Water Cycle in Power Plants

Water Cycle: Condensate extraction -> LP/HP heaters -> Deaerator (removes O2) -> Boiler feed pump (multistage, 300 bar in USC) -> Boiler -> Turbine -> Condenser.

Losses: 1-2% in leaks (evaporation), 5% in blowdown (purity maintenance).



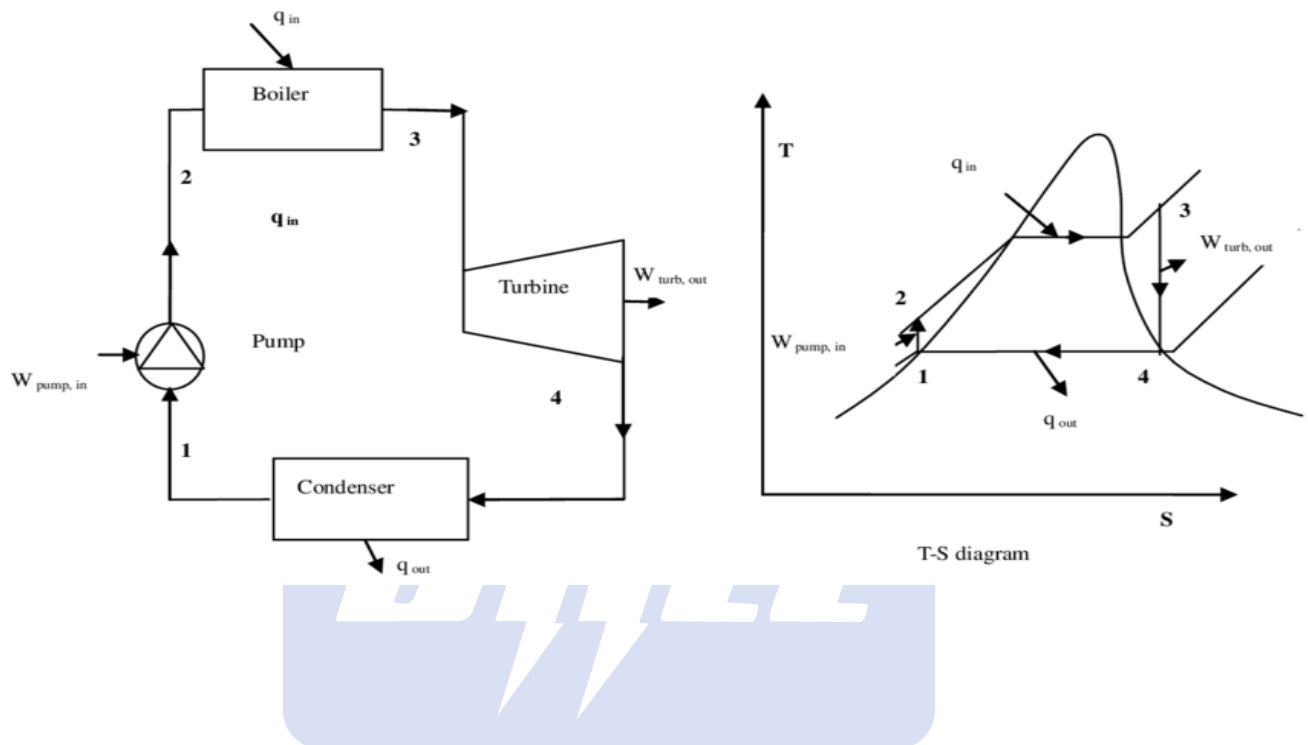
Condenser (0.1 bar, vacuum) -> Extraction Pump -> LP Heaters (100°C) -> Deaerator (removes dissolved gases) -> HP Heaters (200°C) -> Boiler (565°C, 170 bar) -> Turbine.
Arrows with enthalpy rise equations: $h = C_p \times \Delta T$

Analysis: Nag's book emphasizes deaeration to prevent boiler corrosion; Kusile uses vacuum deaerators for 5% efficiency gain.

Heat Cycle in Power Plants

Heat Cycle: Rankine cycle with superheating (565°C), reheat (IPT), regeneration (feedwater heaters).

Losses: 20-30% in condenser (cooling tower heat rejection), 10% in turbine friction (blade losses).



Boiler (heat input, Q_{in}) $\rightarrow$ Turbine (work output, W_t) $\rightarrow$ Condenser (heat rejection, Q_{out}) $\rightarrow$ Pump (work input, W_p) $\rightarrow$ Boiler. Reheat loop at IPT; regeneration via heaters.

Efficiency equation: $\eta = (W_t - W_p) / Q_{in} = 1 - T_{cold}/T_{hot}.$

Kusile's reheat boosts efficiency 3-5%; molten salt recovery for 10-15% gains. Narora's nuclear cycle used similar regeneration but with fission heat.



Waste Heat Recovery and Molten Salt Storage

A Replacement to Chillers?

“Thinking about wasted heat, my research at Narora Nuclear Power Plant came in handy. Molten salt storage fascinated me as a way to harness surplus thermal energy. I got into the weeds: nitrate salt mixes, hot/cold storage, heat exchangers, pumping, control systems—all of it. Learned hands-on about materials engineering as well (Hastelloy, Inconel 625).

Flue gas desulfurization (BHEL’s Korba West STPP) was a rabbit hole, but I hit a block figuring out the best site for heat extraction in thermal plants.”

Waste heat (30-40% loss in exhaust) can be recovered using molten salt storage, inspired from my Narora nuclear study.

Process: Nitrate salt (60% NaNO_3 , 40% KNO_3) stores heat in hot/cold tanks (290–565°C) via heat exchangers, pumped (300 kg/s) for steam generation.

Components: Tanks (SS316, 1,000 m³), Heat Exchangers (Inconel 625, 95% efficiency), Pumps (centrifugal), Controls (AI like Daikin iTM).

Materials: Hastelloy for corrosion (<0.1 mm/year at 565°C).

Challenges: Exact extraction point (post-FGD flue gas, 150–200°C) requires site data—dead end in research, but potential in BHEL’s Korba West STPP.

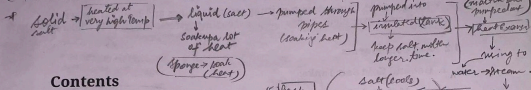
→ realistic data of steam cycle if could be provided.

specific
FC → Condens.
pump

* Electrolysis
H₂

2-4 days - turn off.

Molten Salt Storage for BHEL Thermal Power Plants



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1 Technical Specifications

1.1 System Overview

- **Purpose:** Store excess thermal energy from thermal power plant boilers (coal, gas, or lignite) during off-peak periods and dispatch it for steam generation during peak demand, enhancing load flexibility and efficiency.
- **Heat Source:** Boiler exhaust gases (150 °C to 600 °C for coal, 1000 °C to 1400 °C for gas turbines) or direct steam (500 °C to 600 °C), compatible with molten salt storage temperatures.

1

Nitrate salt $\rightarrow$ Thermal Gaining
Stable $\rightarrow 565^\circ\text{C}$
 $= 1.5 \text{ MJ kg}^{-1}$

- **Storage Medium:** Nitrate salt mixture (60% sodium nitrate, 40% potassium nitrate), stable at 290°C to 355°C , with high thermal capacity ($1.5 \text{ MJ kg}^{-1} \text{ K}^{-1}$ to $2 \text{ MJ kg}^{-1} \text{ K}^{-1}$) and low vapor pressure.
- **Capacity:** Modular units of 100 MW h to 500 MW h thermal storage, scalable to 1000 MW h to 2000 MW h for large plants (e.g., 800 MW supercritical units).
- **Integration:** Coupled with existing steam turbines and heat exchangers, replacing or supplementing traditional cooling systems (e.g., R-134a chillers).

1.2 Key Components

1. Hot and Cold Salt Tanks:

- Material: Stainless steel (SS316) with internal ceramic insulation to resist corrosion at 565°C . (666 K)
- Capacity: 500 m^3 to 2000 m^3 per tank, storing 1000 t to 4000 t of molten salt (e.g. 1000 MWh requires 2000 t at $1.5\text{ MJ kg}^{-1}\text{ K}^{-1}$).
- Design: Double-walled, vacuum-insulated tanks with external heating to maintain salt liquidity (melting point 220°C).

2. Heat Exchangers:

- **Type:** Shell-and-tube, designed for high-temperature steam-to-salt or gas-to-salt heat transfer.
- **Material:** Inconel 623 or Hastelloy for corrosion resistance at 365 °C
- **Efficiency:** 95-98% heat transfer efficiency, based on CSP plant benchmarks (e.g., Solar Two).
Corrosive environment

2 Pumps and Piping

- **Pumps:** Vertical centrifugal pumps for molten salt circulation, rated for 565 °C and 2 MPa to 5 MPa pressure.
- **Piping:** Insulated stainless steel with trace heating to prevent salt solidification.
- **Flow Rate:** 100 kg s⁻¹ to 500 kg s⁻¹, depending on plant capacity (e.g., 400 MW plant requires 300 kg s⁻¹ for 1000 MWh storage).

4. Control Systems:

- **Type:** IoT-enabled PLC with AI optimization, similar to Daikin's IIM, for real-time monitoring of temperature, flow, and energy dispatch.
- **Features:** Predictive maintenance, load forecasting, and integration with plant SCADA systems.
- **Standards:** Compliant with IEC 61850 for power system automation.

5. Emission Control Integration:

- in electrical substitution

3

- **Flue Gas Desulfurization (FGD):** Paired with BHEL's FGD systems (as used in Korba West STPP) to reduce SOx emissions during peak operation.
- **Heat Recovery:** Captures waste heat from FGD exhaust (150 °C to 200 °C) for additional storage, boosting efficiency.

1.3 Performance Metrics

- **Thermal Efficiency:** 99% storage efficiency (based on CSP benchmarks), with 5–7% overall plant efficiency gain (38% to 42–44% for coal plants).
- **Storage Duration:** 4–8 hours at full load (e.g., 500 MW for 5 hours from 1000 MW h storage).
- **Life Cycle:** 20–30 years, with minimal degradation of nitrate salts.
- **Operating Conditions:** Stable at 290 °C to 565 °C, with low maintenance due to non-toxic salt properties under controlled conditions.

1.4 Standards and Compliance

- Material Standards: ASME Section III Division 5 for high-temperature components, IS 16590 for thermal systems in India.
- Environmental Compliance: Aligns with India's HFC phase-down (Kigali Amendment) and MoEFCC emission norms (SOX < 200 mg/m³).
- Safety: Non-flammable salts (A1 classification), with leak detection and containment systems to meet AERB and OSHA standards.

2 Cost-Benefit Analysis

2.1 Capital Costs

- **Storage System:** 5 007million INR (e.g., 500 00700crore INR for 1000 MW h), including tanks, heat exchangers, pumps, and controls.

- **Retrofitting:** 50-100 crore INR for a 500 MW plant, covering heat exchanger modifications and piping upgrades.

- **New Installation:** 700-1000 crore INR for a 2x800 MW plant (e.g., Singrauli STPP), including storage and emission control systems. 1000 cr
- **Funding Support:** 20-30% cost reduction (100-200 crore INR) via PLI scheme & subsidies, leveraging BHEL's Maharatna status.

- **Funding Support:** 20-30% cost reduction (100-200 crore INR) via PLI scheme or MoR subsidies, leveraging BHEL's Maharatna status.

☐ $\rightarrow$ extra energy (heat) $\rightarrow$ stored (Sodium + Potassium) Nitrate storage $\rightarrow$ ☐ Peak load release out.

PP doesn't need to run on high power.



Table: Molten Salt System Detailed Specs

Component	Material	Capacity	Efficiency	Cost (INR crore)	BHEL Application
Tanks	SS316	1,000 m ³ (2,000 tons salt)	99% thermal	2-3	Post-FGD integration
Heat Exchanger	Inconel 625/Hastelloy	N/A	95-98%	1-2	Flue gas heat capture
Pumps	Centrifugal	300 kg/s	N/A	0.5	Circulation in Noida designs

Cost-Benefit:

Capital 5-7 million INR/MWh, savings 15-25 lakh INR/MW/year in fuel (payback 5-7 years).

10-15% CO₂ reduction, aligning with net-zero.

Analysis:

TerraPower Natrium show nuclear applications; thermal adaptation boosts efficiency 5-7%.

Narora used similar storage for nuclear waste heat—BHEL can adapt for flue gas, using FGD (99% SO_x removal in Korba West).

Can Extract post-economizer (300°C).

Hitting the dead end was frustrating, but constant guidance and motivation kept me going.



Combined Cycle Power Plants (CCPPs) and Hydrogen Fuel

When I explored Combined Cycle Power Plants (CCPP), Sachin sir pointed out a key sticking point: constant gas fuel supply. That led me to research hydrogen-based clean fuel, compact generation, and supply at scale for power plants. I tracked what leaders like ACES Delta Hub and Ansaldo Energia are doing, and got excited about new turbine designs like GE Vernova's 9FA and 9HA.02.

CCPPs: Gas turbine (1600°C) + HRSG (steam generation) + steam turbine (565°C), 60% efficiency.

Challenge: Gas supply, discussed with Sachin Sir. Explored hydrogen: Clean fuel, process from electrolysis/SMR/gasification.

Innovations: ACES Delta Hub (storage), Ansaldo Energia (50% H₂-ready turbines), GE Frame 9HA.02 (BHEL license, 300 MW, 1600°C inlet).

Heat Cycle in CCPP: Gas (high T) -> Steam (reheat at IPT, pressure drops HPT 170 bar to LPT 0.1 bar).

We were discussing if Electrolysing and then converting water into Hydrogen makes sense and it already got implemented :- [Hydrogen-Fueled Gas Turbines | GE Vernova](#)

Hydrogen reduces CO₂ 20-30% at 30% blend. BHEL's Palatana CCPP (726 MW) can retrofit with 316L steel for embrittlement.

Hydrogen production specs: Electrolysis (50 MW, 24 tons/day), SMR (90% CCUS), gasification (2 tons/day from biomass).]

Discussion with Sachin Sir on combined cycle Power Plant sparked my research; linking it to Narora's clean energy felt full-circle.



Turbine Systems, Blade Designs, and Piping Insights

I learned the turbine breakdown: High Pressure (HPT), Intermediate Pressure (IPT), and Low Pressure (LPT) turbines, plus the MS line. The abbreviations in piping diagrams started making sense

Each one has a job, and understanding them made reading plant flowcharts a lot easier.

I got deep into turbines—their blade shapes, weights, sizes, and how their design changes based on plant type (HPT, IPT, LPT). Each improvement in blade design, especially for supercritical and ultra-supercritical setups, was fascinating and important for increasing efficiency.

Turbines: HPT (170 bar, 565°C, small titanium blades), IPT (reheat, medium), LPT (vacuum, large, weight 10x HPT blades).

Blade Designs: USC uses 3D curved profiles, TBC (yttria-stabilized zirconia, reduces heat loss 5-10%), materials like titanium for high-speed (1,500 m/s tip velocity).

MS Line: Main Steam line (SA213 T92, 565°C, 170 bar).

Piping Abbreviations :

- PI: Pressure Indicator (local gauge, 0-200 bar).
- PT: Pressure Transmitter (4-20 mA signal to SCADA for remote monitoring).
- PS: Pressure Switch (activates alarm at 150 bar threshold).
- LI: Level Indicator (local sight glass, 0-5 m).
- LT: Level Transmitter (continuous signal for boiler drum).
- LS: Level Switch (high/low alarms for safety shutdown).
- TI: Temperature Indicator (local thermocouple, 0-600°C).
- TT: Temperature Transmitter (remote signal for turbine inlet).
- TS: Temperature Switch (cutoff at 600°C to prevent overheating).

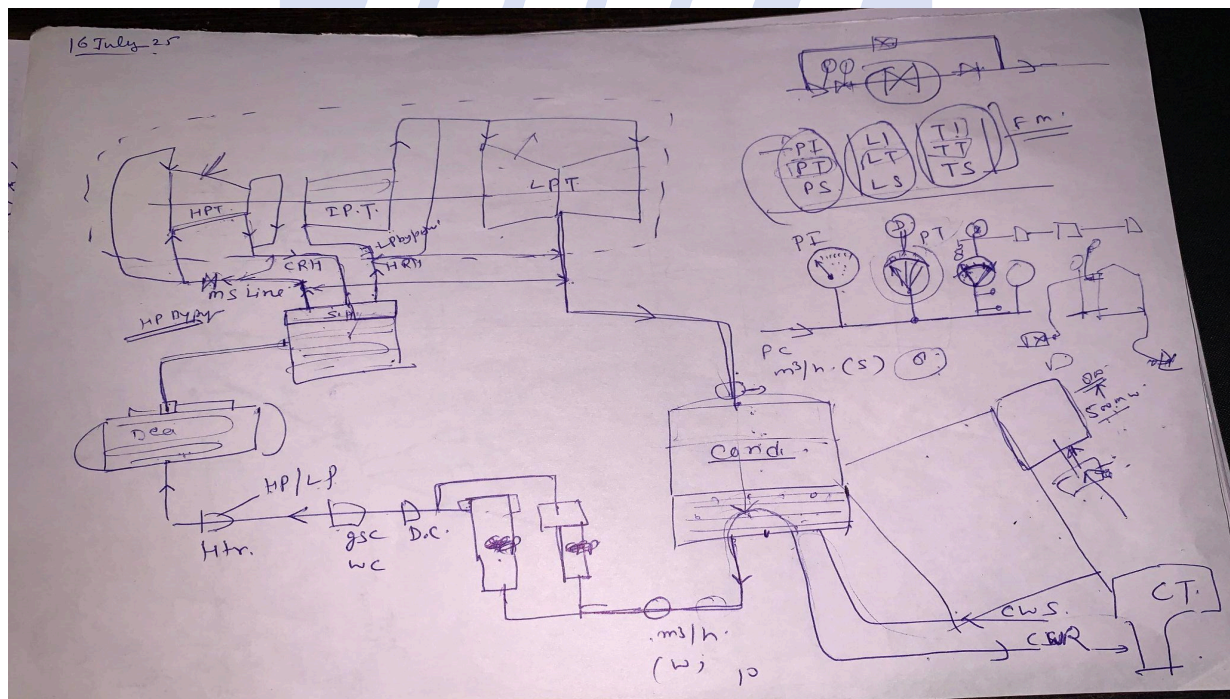
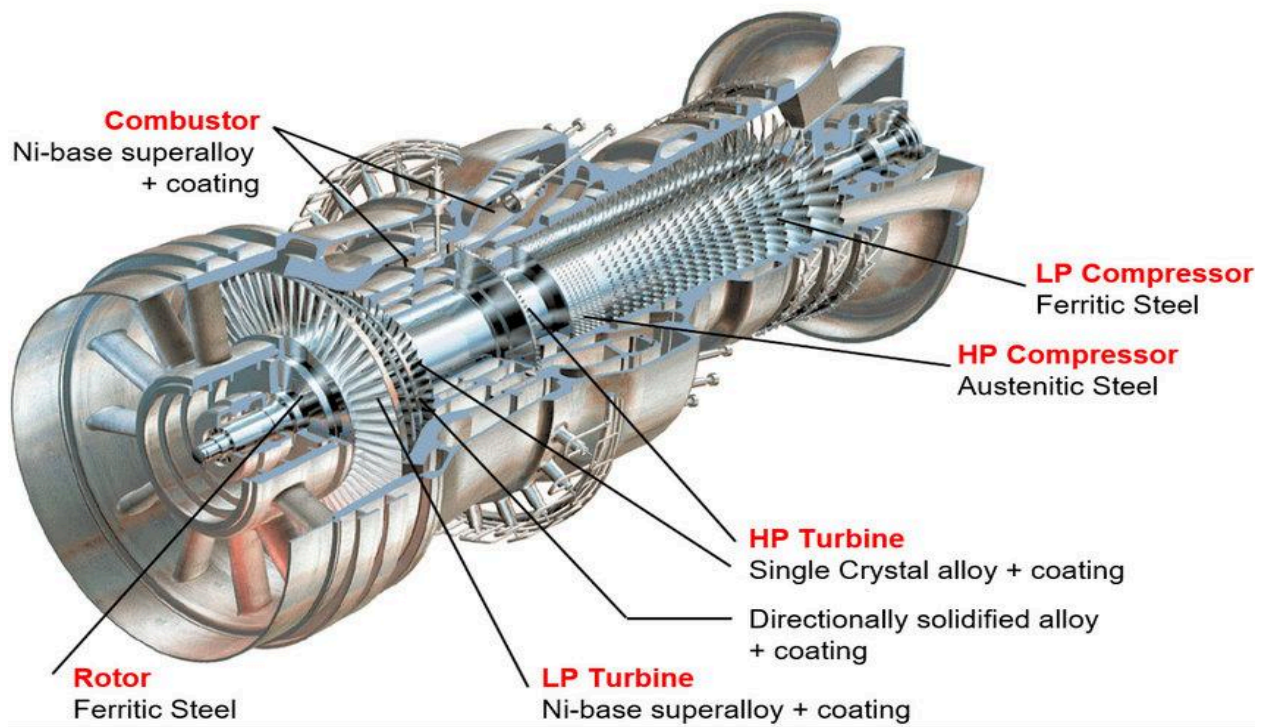




Table: Blade Differences in Plant Types

Plant Type	Blade Size (HPT/LPT)	Material	Aerodynamics	Efficiency Gain	BHEL Change
Subcritical	Small/large	Steel	2D	Baseline	Basic
Supercritic	Smaller/larger	T91	3D curved	3-5%	Thicker coatings
USC	Compact/massive	T92/Inconel	Advanced 3D + TBC	5-10%	Curved profiles reduce drag

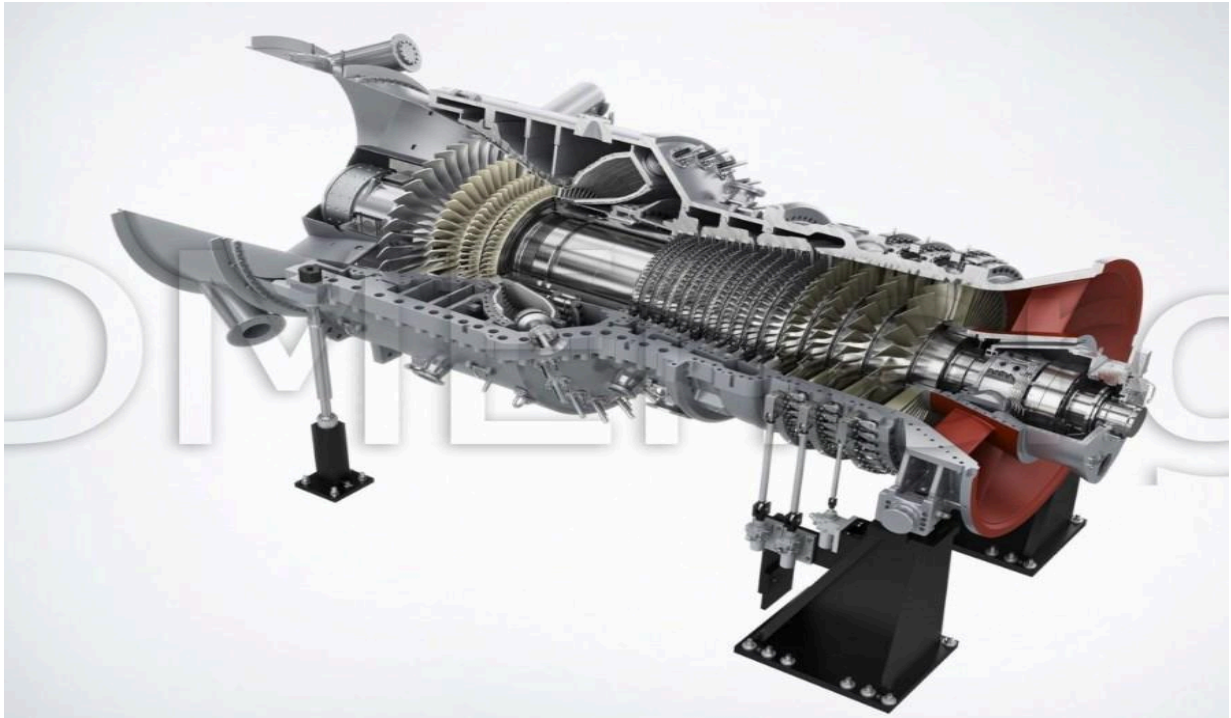
GE 9HA.02 Gas Turbine

- Latest Generation "H-class" Turbine
- Net Output: 571MW (simple cycle, ISO conditions)
- Net Efficiency (LHV): 44.0% simple cycle, up to 64% in combined cycle mode (industry record)
- Ramp Rate: 88MW/min (enables fast response to load changes)
- Startup Time: ~23 minutes (hot start)
- Hydrogen Capability: Can run on fuel blends up to 50% hydrogen, future-ready for more
- Applications: Large combined cycle plants, grid peaking and flexible operation
- Key Feature: Highest output and efficiency in segment, with ultra-low emissions
- Operational Flexibility: Up to 70MW/minute within emissions compliance
- Reference: World's most efficient heavy-duty gas turbine; validated by real plants in Malaysia and China.



GE 9FA Gas Turbine

- Proven F-class Turbine
- Net Output: 288–332MW (depending on model, typically 9F.04)
- Net Efficiency (LHV): >60% in combined cycle mode
- Ramp Rate: Lower than H-class; designed for steady base load but supports cycling
- Hydrogen Capability: Up to 10% blending
- Dry Low NO_x Combustion: Lower emissions than older technologies
- Applications: Combined cycle and CHP (cogeneration), both utility and industrial
- Key Feature: Robust reliability record worldwide (450+ units installed), versatile fuel handling
- Reference: Segment workhorse, widely adopted in 50Hz grids



From Nag's book and Kusile video, USC blades improve efficiency 3-5% via reduced boundary layer losses (Bernoulli's equation: $P + 0.5\rho v^2 + \rho gh = \text{constant}$). BHEL's AUSC (NTPC 800MW) uses Inconel valves for 700°C, differing from subcritical's carbon steel.

Piping insights from Jasjyot helped understand how PT/TT signals prevent failures.

Deciphering abbreviations with Jasjyot felt like cracking a code.

Bus Duct, Transmission, and Transformers

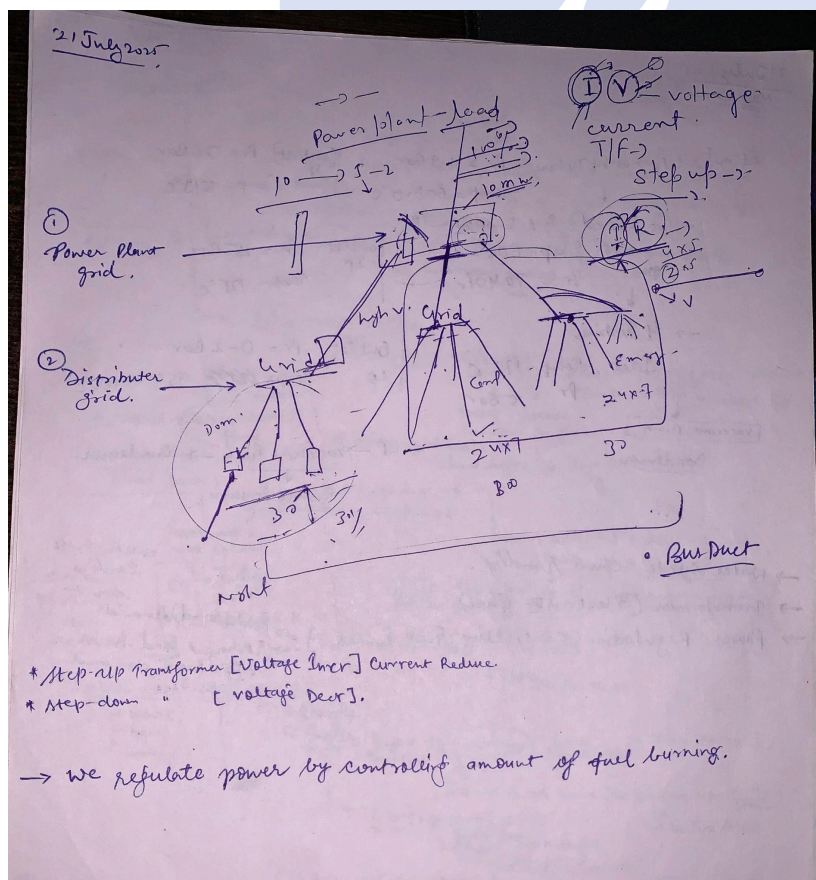
I also picked up the basics about bus ducts, transmission systems, grid distribution, and step-up/step-down transformers—why they matter and how they connect the dots in the power chain.

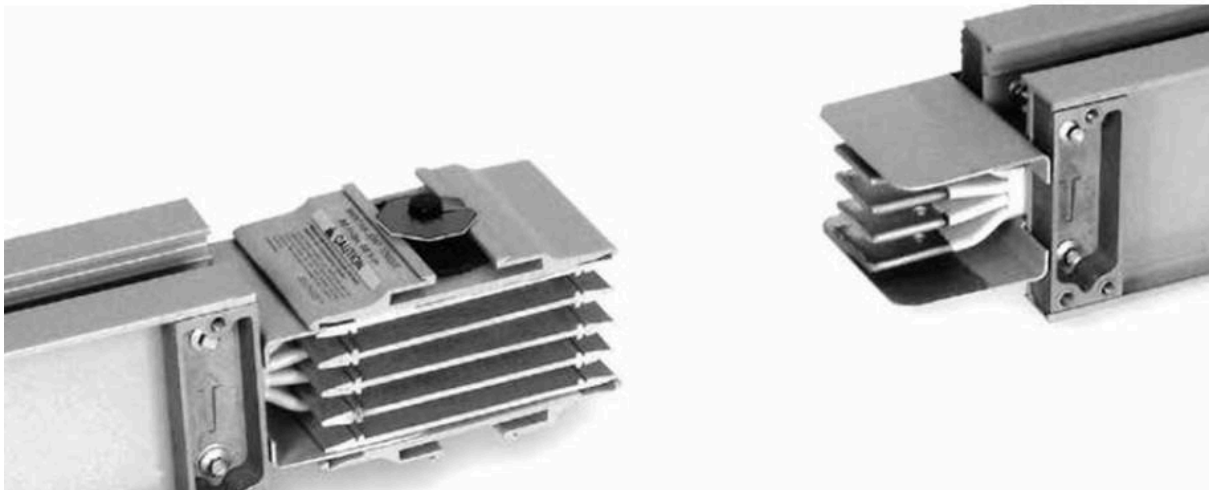
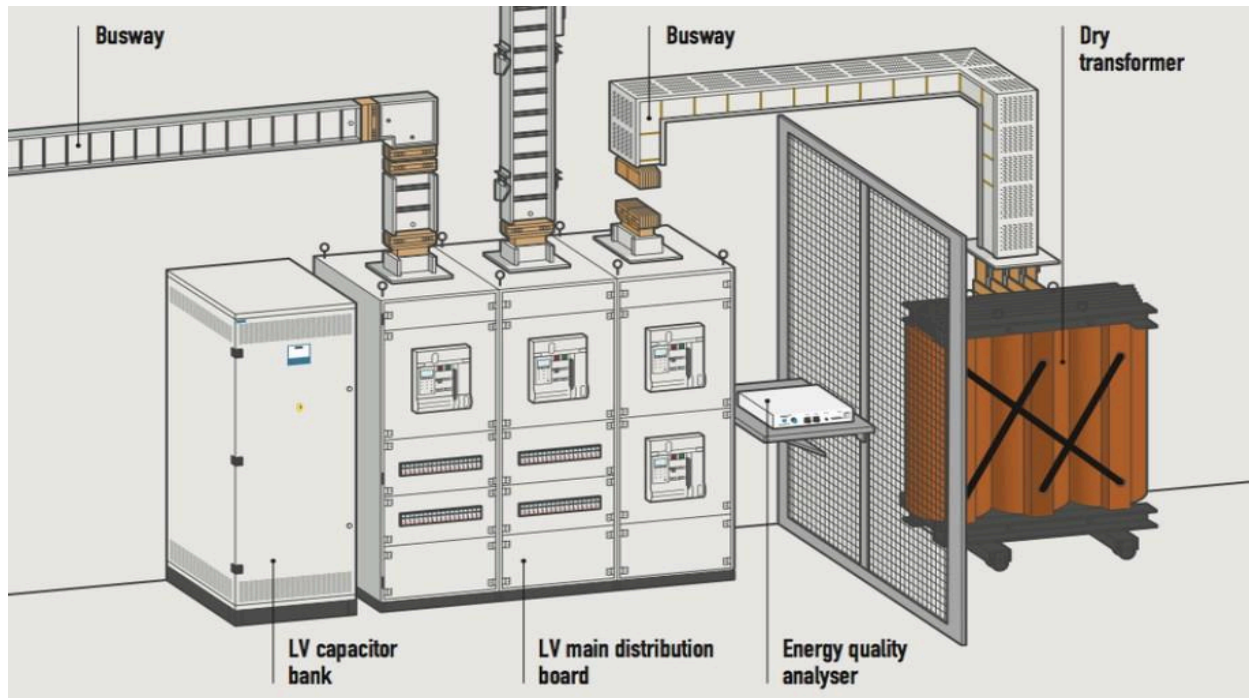
Bus Duct: Enclosed conductors (aluminum/copper, 10-20 kA) for low-loss power transfer from generator to transformer.

Transmission System: Generator (11-22 kV) → Bus Duct → Step-Up Transformer (220-400 kV) → Grid → Step-Down Transformer (distribution).

Step-Up Transformers: Core-type, oil-immersed (ONAN cooling), 99% efficiency, reduce transmission losses (I^2R). - We Decrease the level of Current (Voltage Incr)

Step-Down: Similar, for local use. - We Increase level of Current (Voltage Decr).





Bus ducts cut losses 2-3% vs. cables. BHEL's Ramagundam STPP uses ONAN transformers with hysteresis loss minimization. Narora's transformers had radiation shielding, unlike BHEL's focus on efficiency coatings.

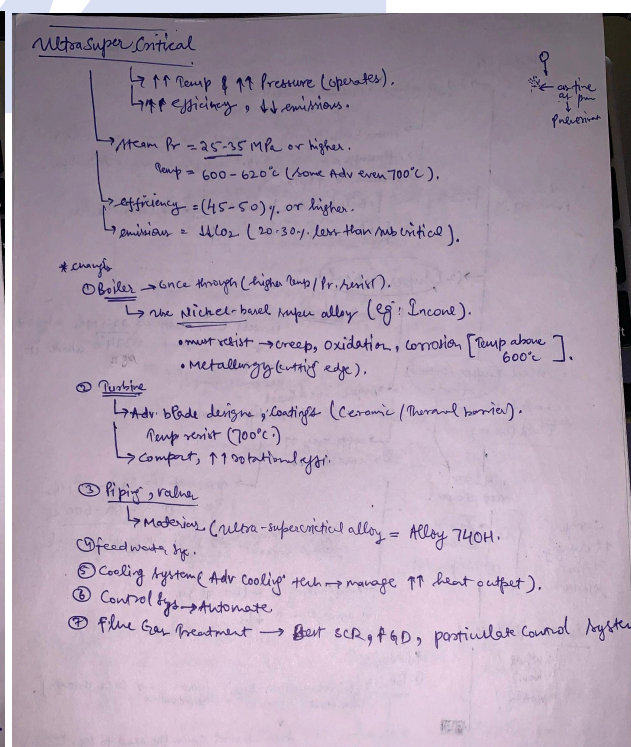
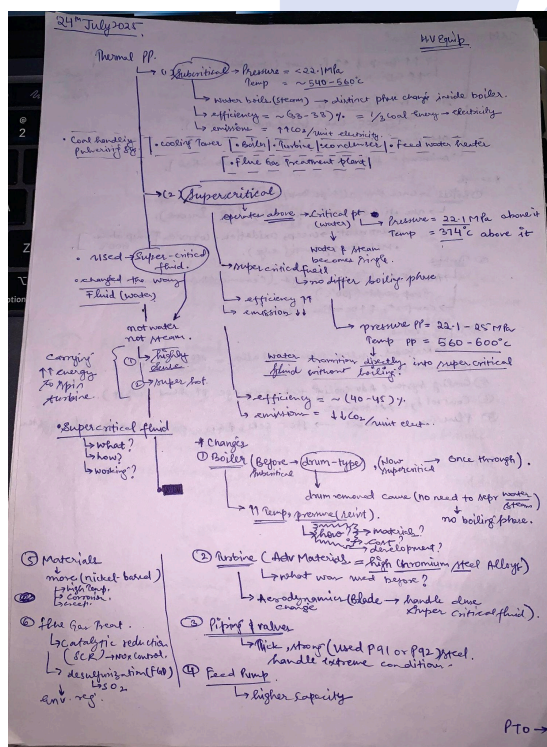
Subcritical, Supercritical, and Ultra-Supercritical Plants

I learned how heat cycle, steam temperature, and pressure change as they move through HPT, IPT, and LPT stages. I made sense of subcritical, supercritical, and ultra-supercritical power plants—how each affects boiler design, turbine construction, piping, valves, pump tech, materials, and emissions treatment.

What stood out was the aerodynamic tweaks to turbine blades for ultra-supercritical plants—the little changes that create efficiency gains at scale. Examples from BHEL's Yerams Thermal PP and NTPC-BHEL's 800 MW AUSC plant were eye-opening.

Detailed comparison:

- Subcritical: <221 bar, <374°C, 32–38% efficiency, carbon steel, basic FGD.
- Supercritical: >221 bar, >374°C, 38–42% efficiency (Yerams Thermal PP), SA213 T91 materials, enhanced FGD/SCR.
- USC: >250 bar, >600°C, 42–48% efficiency (NTPC-BHEL 800MW AUSC), T92/Inconel, advanced SCR, 10-15% less coal.





BHEL Changes in USC:

- Boiler: Once-through, high-pressure feed pumps.
- Turbine: 3D blades with TBC (yttria-stabilized zirconia, reduces heat loss 5-10%, equation: Heat flux $q = k \Delta T / \delta$).
- Piping/Valves: Thicker Inconel 625 for 700°C.
- Flue Gas: FGD (99% SO_x, Korba West), SCR (90% NO_x).

Expanded Plant Comparison

Parameter	Subcritical	Supercritical	Ultra-Supercritical	BHEL Example	Efficiency Impact
Pressure (bar)	<221	>221	>250	Yerams (SC), AUSC 800MW (USC)	Higher P = better η
Temperature (°C)	<374	>374	>600-700	AUSC 700°C	5-10% gain
Efficiency (%)	32–38	38–42	42–48	AUSC 46%	Reduced fuel 10-15%
CO ₂ (kg/kWh)	0.9–1.0	0.8–0.9	0.7–0.8	AUSC lowest	Net-zero alignment
Materials	Carbon steel	SA213 T91	T92/Inconel	USC turbines (3D blades + TBC)	Corrosion resistance



FGD

Basic

Enhanced

Advanced
SCR

Korba West

95-99%
SO_x/NO_x
cut

Analysis - USC cuts CO₂ 10-15%.

BHEL's AUSC uses TBC on blades for 3-5% efficiency, differing from subcritical's simple designs.





Supercritical CO₂ in Turbines

Efficiency though the best interconnection between department.

Supercritical CO₂, Research, and Real-World Demos

My last big topic was researching supercritical CO₂ in next-gen turbines, including what the STEP Nuclear Energy Demo in Texas achieved, and the engineering challenges and solutions they found.

sCO₂ Cycle: Brayton, CO₂ compressed to >73.8 bar (>31°C), heated to 565°C, expanded.

Process Details:

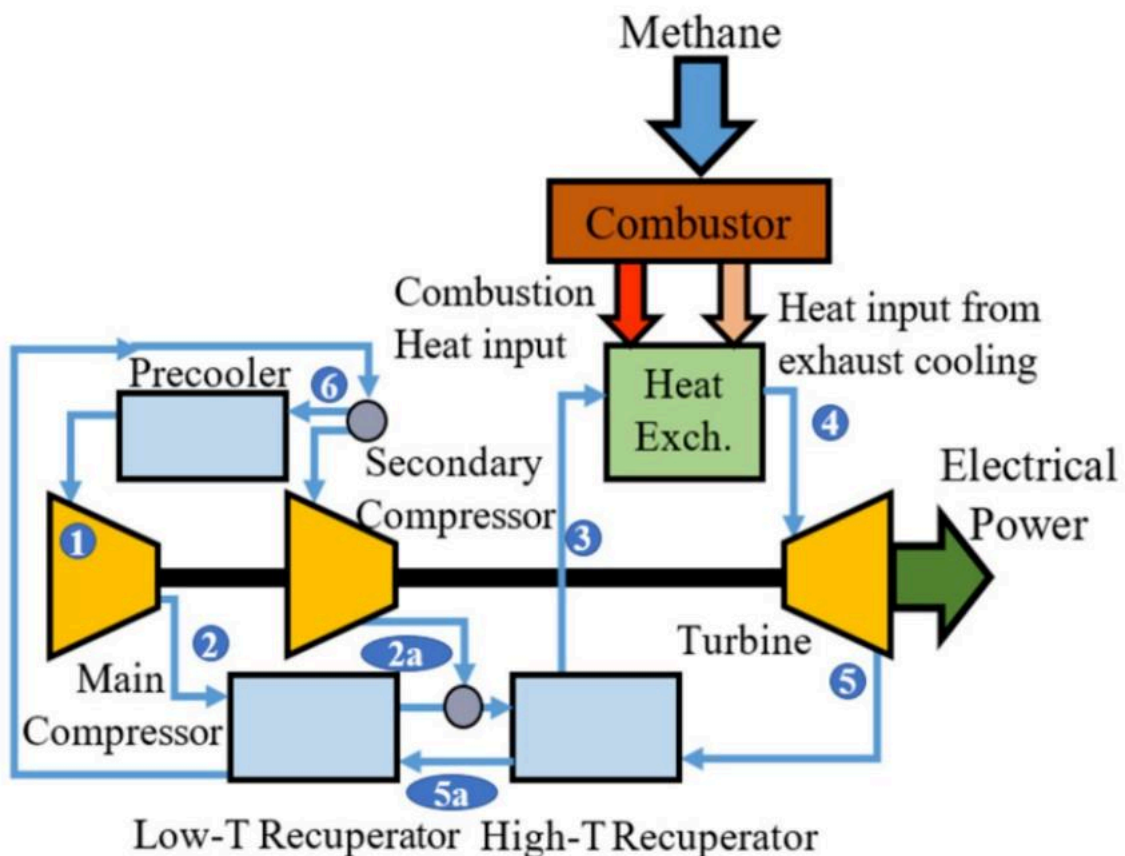
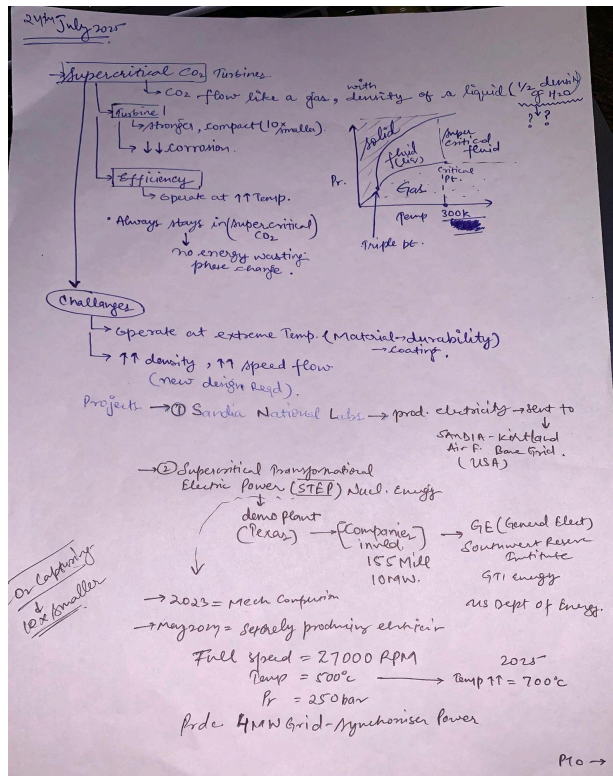
1. Compression: To 300 bar (pump-like efficiency).
2. Heating: In boiler/HRSG ($Q_{in} = m \times C_p \times \Delta T$).
3. Expansion: Turbine work ($W_t = m \times h_{in} - h_{out}$).
4. Cooling: Recompression.

Advantages: 50–55% efficiency, compact (1/10th steam size), 90% less water.

Challenges: High P (corrosion, Inconel 625), seals (leak rate <0.1%).

Higher Density, so specific blades needed.

Reference: STEP Demo Plant, Texas (10 MW, 50% efficiency, GTI Energy).





[STEP Demo: High-efficiency power generation pilot program • GTI Energy](#)

Analysis: (GTI Energy) suggests sCO₂ for BHEL's AUSC; 5-10% efficiency over steam.

Narora's nuclear hybrids could adopt sCO₂ for similar gains, but thermal plants need CO₂-resistant materials like Hastelloy.

sCO₂'s compactness amazed me.





Ending Notes:-

While my internship gets over, learning continues.

- Working on integrating multiple power plants and creating an efficient system.

Gratitude and Acknowledgments:-

- Sachin Sir: For providing insights and helping me learn HVAC, turbines, CCPPs, and more—truly blessed to have your guidance, always my Go-To for any issue. From HVAC to Layouts to Turbine Systems gained so much learning during all this time.
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BHEL and the MAUX Department changed how I see engineering, energy, and teamwork. Every challenge, every conversation, and every late-night learning session contributed to my growth—and I'm proud of both what I learned and the people I learned it with.

This internship at BHEL Noida felt like a family effort, with mentors and peers supporting my growth. Rajender Sir's Hindi lessons added a cultural dimension to my technical learnings.



Conclusion

Key Takeaways:

- Proposed R-32 for new HVAC (household success scaled to industry via Indian manufacturers), R-513A for retrofits, saving 50 lakh INR/plant and meeting 2028 phase-down/net-zero goals.
- Explored molten salt for 5-7% efficiency, hydrogen for CCPPs, sCO₂ for USC—bridging my Narora insights with BHEL's thermal focus.

Recommendations:

- Collaborate with Blue Star/Voltas for R-32 production/installation (industry level, 20-30% cost reduction).
- Invest 20-30 crore INR in molten salt pilots for Korba West.
- Pilot hydrogen in Palatana CCPP (30% blend, 50 crore INR).



Thank you for this opportunity at BHEL Noida!

This internship fueled my passion for sustainable energy, blending my insights with BHEL Noida's thermal expertise. I'm excited to contribute to India's net-zero future!

